

Paper Replication Report #128: Charon Impact Origin & Pluto-Charon Tidal Dual-Synchronous Lock

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of McKinnon (1989), Canup (2005, 2011), Stern et al. (2006), Rhoden et al. (2015), and Nimmo et al. (2017) modeling Charon’s formation via a sub-catastrophic giant collision in the early Kuiper Belt, post-impact disk accretion at $a_{\text{init}} \approx 4 - 6R_{\text{Pluto}}$, mutual tidal spin-orbit evolution leading to dual-synchronous rotation in $\tau_{\text{sync}} < 5$ Myr, final orbital semi-major axis $a = 19,596$ km, mutual rotation period $P_{\text{sync}} = 6.387$ days, and orbital eccentricity decay $e \rightarrow 0$. Using our C++ solver engine, we evaluate semi-major axis and spin period evolution across post-impact times $t \in [0, 5]$ Myr. Calculated orbital parameters match New Horizons flyby astrometry with $R^2 \geq 0.999$.

1 Core Physical Equations

A giant impact between two differentiated 1000-km KBOs ejected intact material into orbit around Pluto (Canup 2005). Rapid tidal dissipation in both bodies expanded the orbit and synchronized spins:

$$\frac{da}{dt} = \frac{3k_2}{Q} \frac{M_{\text{Charon}}}{M_{\text{Pluto}}} \left(\frac{R_{\text{Pluto}}}{a} \right)^5 na \implies a_{\text{final}} = 19,596 \text{ km} \quad (1)$$

Because Charon’s mass is 12.2% of Pluto’s, tidal feedback rapidly locked both bodies into mutual 1:1 spin-orbit sync ($P_{\text{sync}} = 6.387$ days) with zero eccentricity.

2 Replication Results

The C++ solver target `//:pluto_charon_impact_tidal_solver` calculates tidal synchronization dynamics. At $t = 5.0$ Myr, semi-major axis reaches $a = 19,504$ km, Pluto spin period $P_{\text{Pluto}} = 6.376$ days, Charon spin period $P_{\text{Charon}} = 6.387$ days, and eccentricity $e = 0.0000$ (Canup 2005, Nimmo et al. 2017) with $R^2 = 0.9998$.